Final FD worker and rho: bounded build-binding plan

Task T-1558. 3 November 2026.

Route	Diagnostic FD applied; root integration pending
Build proposal	1200s total; static worker + two host mappers
Binding	Source / protocol / executable / image joins
Admission	No execution; conditional budget only

Ordinary software evidence only. Source/build planning; no build or runtime. T-1558.

Prepared a concrete source/build plan for optimized static musl ecbench and host analyze_pair/analyze_cohort mappers. The same worker executable contains the enrolled candidate and matched rho/rho_AA methods; method/configuration/profile pins distinguish the arms. Original1260 targets and gate trialcaps remain unchanged. No build or runtime operation was executed.

The critical source dependency is T1554: diagnostic Some(sink) selects pencil_progress_native, whose FD implementation passed6,398 full SieveResult pairs and is now applied as exact canonical SHA87f3581c. Publication and final root integration testing remain required. Canonical None selects native_solver. The new GNU unoptimized library and old musl runtime binary establish neither the final route nor final worker identity. Current Cargo manifest/lock hashes differ from the old static build, so compatibility is not assumed.

The intended build is a single1200-second offline sequence: worker up to1020s, both mappers up to120s, terminal cleanup/hash/ELF reserve60s. Hard8GiB address space and1GiB per file; proposed enforced2CPU/8GiB aggregate boundary must be independently specified before release. Private selected-cache seed is capped2GiB, private registry2GiB, entire staging monitored8GiB with12GiB free required and stop below4GiB. Only six dedicated prior-cache directories were sized, totaling1,448,828KiB; no broad cache inventory or mutation occurred.

Final joins include published source/closure, root tests and reviews, actual worker and mapper binaries, exact finite nine-category mapping, prospective full/canonical protocol digests, image/source/executable readback and real controller ownership pins. Old Q0-Q4 binds image3a2bb98a and workerd0e2d142 only. A proposed later600s assembly plus300s new-image qualification remains separately unadmitted. Postrun CLI0adc4824 is distinct from the worker and still requires independently authenticated caller pins.

Progress diagnostics do not currently supply canonical optional Event::Candidate relation traces. This plan neither adds tracing nor claims a full retained relation stream. Preserve the separate target-information lane and matched timing/A-A profiles; report observer, traversal, digest and serialization overhead in their own units.

The saved59USD reserve includes4USD prior/completed-attempt allocation and55USD shared/delay encumbrance. A proposed2USD build/assembly/qualification/storage suballocation is conditional on coordinator release inside that55, leaving53; funding is presently false. Actual costs are unknown. Historical9.87USD headroom is not used. Root reported unchanged delayed budget values near22:00, which do not provide new spendable confirmation. No campaign debit was made.

commands.json and cache-plan.json contain concrete argv, bounded copy scope and deliberately null final bindings. These are a reviewable preparation recipe, not an executable release. Source observations and binding-matrix.csv identify the remaining joins. Fresh budget, capacity/ownership and exact source checks precede any cloud step.